Prepareation

```
import requests
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")

%matplotlib inline

API_KEY = 'jnl-JWIkbXDZ5fARfOHsjr6VNXIhE7cWVmGShW7OsIN8KDiUOMZfrlQ8wOnsq_D8ofubFtBrOVWdO2babZgmMNEd12pRmdLovzrML9YgjbTCVKPNuLGK9V2eHUYVZ3Yx'
HEADERS = {'Authorization':f'Bearer {API_KEY}'}
```

Data From Related Coffee Stores

```
# Yelp API endpoint
url = "https://api.yelp.com/v3/businesses/search"
```

Blue Bottle Coffee on Manhattan

```
# Parameters for the API request
params = {
    'term':'Blue Bottle Coffee',
    'location': 'New York City',
    'categories': 'Coffee',
    'limit': 50
}


# Make the request to the Yelp API
response = requests.get(url, headers=HEADERS, params=params)

# Parse the response JSON
data = response.json()

# Check if 'businesses' key exists in the response
if 'businesses' in data:
    businesses = data['businesses']

    # Extract business information from JSON
    store_list = []
    for business in businesses:
        store_info = {
            "Name": business.get("name"),
            "Address": ", ".join(business["location"]["display_address"]),
            "Latitude": business["coordinates"]["latitude"],
            "Longitude": business["coordinates"]["longitude"],
            "Zip Code": business["location"].get("zip_code", "N/A"),
            "Rating": business.get("rating"),
            "Review Count": business.get("review_count"),
            "Price": business.get("price", "N/A"),
        }
        store_list.append(store_info)

    # Create a DataFrame from the extracted data
    df = pd.DataFrame(store_list)

df.head()
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price
0	Blue Bottle Coffee	22 Broad St, New York, NY 10005	40.706510	-74.01123	10005	4.3	22	N/A
1	Blue Bottle Coffee	396 Broadway, New York, NY 10013	40.718480	-74.00248	10013	4.1	37	$$
2	Blue Bottle Coffee - New York	150 Greenwich St, New York, NY 10007	40.710715	-74.01233	10007	4.2	175	$$
3	Blue Bottle Coffee - Dean St	85 Dean St, Brooklyn, NY 11201	40.687360	-73.98977	11201	4.3	152	$$
4	Blue Bottle Coffee	408 Greenwich St, New York, NY 10013	40.721594	-74.01001	10013	3.4	8	N/A

```
df.dtypes
```

```
Name             object
Address          object
Latitude        float64
Longitude       float64
Zip Code         object
Rating          float64
Review Count      int64
Price            object
dtype: object
```

```
# filter out the rows where the store name is not "Blue Bottle Coffee" and ZIP codes are outside Manhattan's range (10001 - 10282)
df['Zip Code'] = df['Zip Code'].astype(int)
df_blue = df[
        (df['Name'].str.contains("Blue Bottle Coffee")) &
        (df['Zip Code'].between(10001, 10282))
]
df_blue
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price
0	Blue Bottle Coffee	22 Broad St, New York, NY 10005	40.706510	-74.011230	10005	4.3	22	N/A
1	Blue Bottle Coffee	396 Broadway, New York, NY 10013	40.718480	-74.002480	10013	4.1	37	$$
2	Blue Bottle Coffee - New York	150 Greenwich St, New York, NY 10007	40.710715	-74.012330	10007	4.2	175	$$
4	Blue Bottle Coffee	408 Greenwich St, New York, NY 10013	40.721594	-74.010010	10013	3.4	8	N/A
6	Blue Bottle Coffee	101 University Pl, New York, NY 10003	40.734030	-73.992520	10003	4.1	62	$
8	Blue Bottle Coffee	1227 Broadway, New York, NY 10001	40.746621	-73.988791	10001	3.9	11	N/A
9	Blue Bottle Coffee	1 Pennsylvania Plaza, Ste 135, New York, NY 10119	40.751186	-73.992120	10119	3.3	8	N/A
10	Blue Bottle Coffee	441 8th Ave, Ste 68, New York, NY 10001	40.751537	-73.995835	10001	3.7	20	N/A
11	Blue Bottle Coffee - Grand Central Place	60 E 42nd St, Ste 140, New York, NY 10165	40.752060	-73.978856	10165	3.8	54	$$
12	Blue Bottle Coffee - Hudson Yards	20 Hudson Yards, Ste 228, New York, NY 10001	40.753119	-74.001823	10001	3.5	97	$$
13	Blue Bottle Coffee - Bryant Park	54 W 40th St, New York, NY 10018	40.753064	-73.984223	10018	4.0	331	$$
14	Blue Bottle Coffee	54 W 40th St, New York, NY 10018	40.753080	-73.984160	10018	3.7	9	N/A
15	Blue Bottle Coffee	1 Rockefeller Plz, Ste N, New York, NY 10020	40.758090	-73.979010	10020	4.2	6	N/A
16	Blue Bottle Coffee	10 E 53rd St, New York, NY 10022	40.759618	-73.975489	10022	3.9	86	$$
17	Blue Bottle Coffee - Midtown North	1345 Avenue Of The Americas, New York, NY 10105	40.763006	-73.979072	10105	4.1	20	N/A
18	Blue Bottle Coffee - Amsterdam Ave	279 Amsterdam Ave, New York, NY 10023	40.779306	-73.980923	10023	3.8	56	$$$

```
df_blue.to_csv("blue_bottle_coffee_manhattan.csv", index=False)
print("CSV file saved successfully.")
```

```
CSV file saved successfully.
```

Joe Coffee Company on Manhattan

```
params_2 = {
    'term':'Joe Coffee Company',
    'location': 'New York City',
    'categories': 'Coffee',
    'limit':50
}


# Make the request to the Yelp API
response_2 = requests.get(url, headers=HEADERS, params=params_2)

# Parse the response JSON
data_2 = response_2.json()

# Check if 'businesses' key exists in the response
if 'businesses' in data_2:
    businesses = data_2['businesses']

    # Extract business information from JSON
    store_list_2 = []
    for business in businesses:
        store_info_2 = {
            "Name": business.get("name"),
            "Address": ", ".join(business["location"]["display_address"]),
            "Latitude": business["coordinates"]["latitude"],
            "Longitude": business["coordinates"]["longitude"],
            "Zip Code": business["location"].get("zip_code", "N/A"),
            "Rating": business.get("rating"),
            "Review Count": business.get("review_count"),
            "Price": business.get("price", "N/A")
        }
        store_list_2.append(store_info_2)

    # Create a DataFrame from the extracted data
    df_2 = pd.DataFrame(store_list_2)

df_2.head()
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price
0	Joe Coffee Company	102 Hicks St, Brooklyn, NY 11201	40.698673	-73.994358	11201	3.8	78	$$
1	Joe Coffee Company	141 Waverly Pl, New York, NY 10014	40.733297	-74.000566	10014	4.1	380	$
2	Joe Coffee Company	9 E 13th St, New York, NY 10003	40.735171	-73.993262	10003	3.9	359	$$
3	Joe Coffee Company	185 Greenwich St, LL3110, New York, NY 10007	40.712628	-74.012820	10007	4.0	29	$$
4	Joe Coffee Company	7 South St Slip, New York, NY 10004	40.701345	-74.011612	10004	3.4	18	N/A

```
df_2.dtypes
```

```
Name             object
Address          object
Latitude        float64
Longitude       float64
Zip Code         object
Rating          float64
Review Count      int64
Price            object
dtype: object
```

```
# filter out the rows where the store name is not "Blue Bottle Coffee" and ZIP codes are outside Manhattan's range (10001 - 10282)
df_2['Zip Code'] = df['Zip Code'].astype(int)
df_joe = df_2[
    (df_2['Name']=='Joe Coffee Company')&
    (df_2['Address'].str.contains("New York, NY")) &
    (df_2['Zip Code'].between(10001, 10282))
]
df_joe
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price
1	Joe Coffee Company	141 Waverly Pl, New York, NY 10014	40.733297	-74.000566	10013	4.1	380	$
2	Joe Coffee Company	9 E 13th St, New York, NY 10003	40.735171	-73.993262	10007	3.9	359	$$
4	Joe Coffee Company	7 South St Slip, New York, NY 10004	40.701345	-74.011612	10013	3.4	18	N/A
6	Joe Coffee Company	514 Columbus Ave, New York, NY 10024	40.785768	-73.972978	10003	4.0	396	$
8	Joe Coffee Company	187 Columbus Ave, New York, NY 10023	40.775080	-73.980367	10001	4.0	143	$
10	Joe Coffee Company	1045 Lexington Ave, New York, NY 10021	40.772044	-73.960805	10001	4.1	261	$
11	Joe Coffee Company	550 W 120th St, New York, NY 10027	40.810050	-73.962020	10165	3.8	216	$$
12	Joe Coffee Company	55 W 40th St, Ste 9, Bryant Park, New York, NY...	40.753458	-73.984319	10001	3.3	26	$$
13	Joe Coffee Company	105 E 42nd St Grand Central Dining Concourse, ...	40.752731	-73.977233	10018	4.1	9	N/A
24	Joe Coffee Company	305 Dodge Hall, 305 Dodge Hall, Upper Campus L...	40.810686	-73.961453	10019	5.0	6	N/A
31	Joe Coffee Company	2950 Broadway Pulitzer Hall, New York, NY 10027	40.807631	-73.963733	10007	3.9	8	N/A

```
import re

# Define a function to extract ZIP code from the address using regex
def extract_zip(address):
    match = re.search(r'\b\d{5}\b', address)
    if match:
        return match.group(0)
    else:
        return None

# Apply the function to the Address column and update the Zip Code column
df_joe['Zip Code'] = df_joe['Address'].apply(extract_zip)
df_joe
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price
1	Joe Coffee Company	141 Waverly Pl, New York, NY 10014	40.733297	-74.000566	10014	4.1	380	$
2	Joe Coffee Company	9 E 13th St, New York, NY 10003	40.735171	-73.993262	10003	3.9	359	$$
4	Joe Coffee Company	7 South St Slip, New York, NY 10004	40.701345	-74.011612	10004	3.4	18	N/A
6	Joe Coffee Company	514 Columbus Ave, New York, NY 10024	40.785768	-73.972978	10024	4.0	396	$
8	Joe Coffee Company	187 Columbus Ave, New York, NY 10023	40.775080	-73.980367	10023	4.0	143	$
10	Joe Coffee Company	1045 Lexington Ave, New York, NY 10021	40.772044	-73.960805	10021	4.1	261	$
11	Joe Coffee Company	550 W 120th St, New York, NY 10027	40.810050	-73.962020	10027	3.8	216	$$
12	Joe Coffee Company	55 W 40th St, Ste 9, Bryant Park, New York, NY...	40.753458	-73.984319	10018	3.3	26	$$
13	Joe Coffee Company	105 E 42nd St Grand Central Dining Concourse, ...	40.752731	-73.977233	10017	4.1	9	N/A
24	Joe Coffee Company	305 Dodge Hall, 305 Dodge Hall, Upper Campus L...	40.810686	-73.961453	10027	5.0	6	N/A
31	Joe Coffee Company	2950 Broadway Pulitzer Hall, New York, NY 10027	40.807631	-73.963733	10027	3.9	8	N/A

```
df_joe.to_csv("joe_coffee_company_manhattan.csv", index=False)
print("CSV file saved successfully.")
```

```
CSV file saved successfully.
```

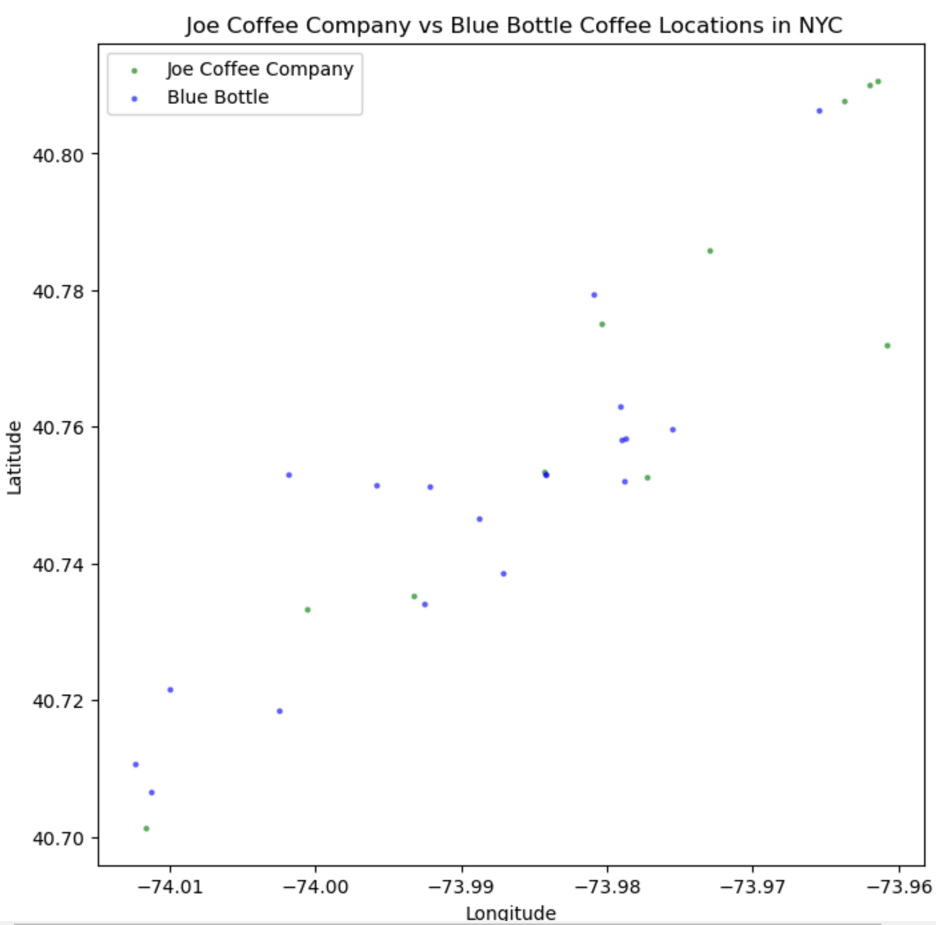
Visulization & Anlysis

Blue Bottle vs Joe Coffee on scatter plot

```
print(f"Joe Coffee Company: {df_joe['Longitude'].shape}, {df_joe['Latitude'].shape}")
print(f"Blue Bottle: {df_blue['Longitude'].shape}, {df_blue['Latitude'].shape}")
```

```
Joe Coffee Company: (11,), (11,)
Blue Bottle: (19,), (19,)
```

```
df_joe['Latitude'] = df_joe['Latitude'].astype(float)
df_joe['Longitude'] = df_joe['Longitude'].astype(float)

df_blue['Latitude'] = df_blue['Latitude'].astype(float)
df_blue['Longitude'] = df_blue['Longitude'].astype(float)


# Plot Starbucks and Blue Bottle stores on the same scatter plot
plt.figure(figsize=(8, 8))

# Scatter plot for Joe locations
plt.scatter(
        df_joe['Longitude'], df_joe['Latitude'],
        s=5, alpha=0.5, color='green', label='Joe Coffee Company'
)

# Scatter plot for Blue Bottle locations
plt.scatter(
        df_blue['Longitude'], df_blue['Latitude'],
        s=5, alpha=0.5, color='blue', label='Blue Bottle'
)


# Add title and labels
plt.title(' Joe Coffee Company vs Blue Bottle Coffee Locations in NYC')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.legend()

# Show the plot
plt.show()
```



Calculation Analysis

```
df_blue.describe()
```

	Latitude	Longitude	Zip Code	Rating	Review Count
count	19.000000	19.000000	19.000000	19.000000	19.000000
mean	40.748169	-73.989480	10031.000000	3.826316	59.473684
std	0.023615	0.013232	45.802717	0.317704	77.639744
min	40.706510	-74.012330	10001.000000	3.200000	6.000000
25%	40.736311	-73.998829	10006.000000	3.650000	15.500000
50%	40.752060	-73.987138	10018.000000	3.900000	37.000000
75%	40.758152	-73.979041	10022.500000	4.100000	59.000000
max	40.806390	-73.965420	10165.000000	4.300000	331.000000

```
df_joe.describe()
```

	Latitude	Longitude	Rating	Review Count
count	11.000000	11.000000	11.000000	11.000000
mean	40.767024	-73.978941	3.963636	165.636364
std	0.035640	0.017210	0.438800	162.700506
min	40.701345	-74.011612	3.300000	6.000000
25%	40.743951	-73.988791	3.850000	13.500000
50%	40.772044	-73.977233	4.000000	143.000000
75%	40.796700	-73.962877	4.100000	310.000000
max	40.810686	-73.960805	5.000000	396.000000

```
# Extracting the mean rating from the summary statistics
mean_rating_blue = df_blue['Rating'].mean()
mean_rating_joe = df_joe['Rating'].mean()

# Comparison using if-else
if mean_rating_blue > mean_rating_joe:
    print("Blue Bottle Coffee has a higher average rating than Joe Coffee Company.")
elif mean_rating_blue < mean_rating_joe:
    print("Joe Coffee Company has a higher average rating than Blue Bottle Coffee.")
else:
    print("Both Blue Bottle Coffee and Joe Coffee Company have the same average rating.")

# Optional: Print the actual mean ratings for reference
print(f"Blue Bottle Coffee Average Rating: {mean_rating_blue:.2f}")
print(f"Joe Coffee Company Average Rating: {mean_rating_joe:.2f}")
```

```
Joe Coffee Company has a higher average rating than Blue Bottle Coffee.
Blue Bottle Coffee Average Rating: 3.83
Joe Coffee Company Average Rating: 3.96
```

Clustering Analysis & Comparison with household income

```
df_blue['Zip Code'].unique()
```

```
array([10005, 10013, 10007, 10003, 10001, 10119, 10165, 10018, 10020,
       10022, 10105, 10023, 10025, 10010])
```

```
blue_by_zip = df_blue.groupby('Zip Code').size().reset_index(name='Store Count')
blue_by_zip
```

	Zip Code	Store Count
0	10001	3
1	10003	1
2	10005	1
3	10007	1
4	10010	1
5	10013	2
6	10018	2
7	10020	2
8	10022	1
9	10023	1
10	10025	1
11	10105	1
12	10119	1
13	10165	1

```
df_joe['Zip Code'].unique()
```

```
array(['10014', '10003', '10004', '10024', '10023', '10021', '10027',
       '10018', '10017'], dtype=object)
```

```
joe_by_zip = df_joe.groupby('Zip Code').size().reset_index(name='Store Count')
joe_by_zip
```

	Zip Code	Store Count
0	10003	1
1	10004	1
2	10014	1
3	10017	1
4	10018	1
5	10021	1
6	10023	1
7	10024	1
8	10027	3

```
#import income data to facilitate analysis
```

```
# Load the dataset
df_income = pd.read_excel(r"C:\Users\ALIENWARE\Desktop\Urban Data Informatics\PSet 02\Median household income.xlsx")
df_income.head(2)
```

	ZipCode	Median_household_income
0	10001	101409
1	10002	37093

```
# Ensure both 'Zip Code' columns are in the same format (as string)
df_blue['Zip Code'] = df_blue['Zip Code'].astype(str)
df_income['ZipCode'] = df_income['ZipCode'].astype(str)

# Merge the datasets based on 'Zip Code'
df_merged_blue = pd.merge(df_blue, df_income, left_on='Zip Code', right_on='ZipCode', how='inner')

# Drop unnecessary 'ZipCode' column from income data after merging
df_merged_blue.drop(columns=['ZipCode'], inplace=True)
```

```
# Save the merged DataFrame to a new CSV
df_merged_blue.to_csv("blue_bottle_with_income.csv", index=False)

df_merged_blue
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price	Median_household_income
0	Blue Bottle Coffee	22 Broad St, New York, NY 10005	40.706510	-74.011230	10005	4.3	22	N/A	197188
1	Blue Bottle Coffee	396 Broadway, New York, NY 10013	40.718480	-74.002480	10013	4.1	37	$$	137572
2	Blue Bottle Coffee - New York	150 Greenwich St, New York, NY 10007	40.710715	-74.012330	10007	4.2	175	$$	250001
3	Blue Bottle Coffee	408 Greenwich St, New York, NY 10013	40.721594	-74.010010	10013	3.4	8	N/A	137572
4	Blue Bottle Coffee	101 University Pl, New York, NY 10003	40.734030	-73.992520	10003	4.1	62	$	137533
5	Blue Bottle Coffee	1227 Broadway, New York, NY 10001	40.746621	-73.988791	10001	3.9	11	N/A	101409
6	Blue Bottle Coffee	441 8th Ave, Ste 68, New York, NY 10001	40.751537	-73.995835	10001	3.7	20	N/A	101409
7	Blue Bottle Coffee - Hudson Yards	20 Hudson Yards, Ste 228, New York, NY 10001	40.753119	-74.001823	10001	3.5	97	$$	101409
8	Blue Bottle Coffee - Bryant Park	54 W 40th St, New York, NY 10018	40.753064	-73.984223	10018	4.0	331	$$	130457

```
df_merged_blue.describe()
```

	Latitude	Longitude	Rating	Review Count	Median_household_income
count	14.000000	14.000000	14.000000	14.000000	14.000000
mean	40.745190	-73.992312	3.814286	70.500000	139460.428571
std	0.026990	0.014102	0.318306	87.566898	40875.414043
min	40.706510	-74.012330	3.200000	8.000000	101409.000000
25%	40.724703	-74.002316	3.625000	20.500000	110194.250000
50%	40.749079	-73.990656	3.850000	40.500000	133872.000000
75%	40.753109	-73.984176	4.075000	80.000000	137572.000000
max	40.806390	-73.965420	4.300000	331.000000	250001.000000

```
# Ensure both 'Zip Code' columns are in the same format (as string)
df_joe['Zip Code'] = df_joe['Zip Code'].astype(str)
df_income['ZipCode'] = df_income['ZipCode'].astype(str)

# Merge the datasets based on 'Zip Code'
df_merged_joe = pd.merge(df_joe, df_income, left_on='Zip Code', right_on='ZipCode', how='inner')

# Drop unnecessary 'ZipCode' column from income data after merging
df_merged_joe.drop(columns=['ZipCode'], inplace=True)

# Save the merged DataFrame to a new CSV
df_merged_joe.to_csv("joe_coffee_with_income.csv", index=False)

df_merged_joe
```

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price	Median_household_income
0	Joe Coffee Company	141 Waverly Pl, New York, NY 10014	40.733297	-74.000566	10014	4.1	380	$	136559
1	Joe Coffee Company	9 E 13th St, New York, NY 10003	40.735171	-73.993262	10003	3.9	359	$$	137533

```
df_merged_joe.describe()
```

	Latitude	Longitude	Rating	Review Count	Median_household_income
count	11.000000	11.000000	11.000000	11.000000	11.000000
mean	40.767024	-73.978941	3.963636	165.636364	123393.545455
std	0.035640	0.017210	0.438800	162.700506	47991.689988
min	40.701345	-74.011612	3.300000	6.000000	58435.000000
25%	40.743951	-73.988791	3.850000	13.500000	94446.000000
50%	40.772044	-73.977233	4.000000	143.000000	136109.000000
75%	40.796700	-73.962877	4.100000	310.000000	137046.000000
max	40.810686	-73.960805	5.000000	396.000000	216017.000000

```
# Extracting the mean rating from the summary statistics
median_income_blue = df_merged_blue['Median_household_income'].mean()
median_income_joe = df_merged_joe['Median_household_income'].mean()

# Comparison using if-else
if median_income_blue > median_income_joe:
    print("Blue Bottle Coffee stores are located in areas with higher average household income.")
elif median_income_blue < median_income_joe:
    print("Joe Coffee Company stores are located in areas with higher average household income.")
else:
    print("Both Blue Bottle Coffee and Joe Coffee Company stores are located in areas with the same average household income.")

# Optional: Print the actual mean incomes for reference
print(f"Blue Bottle Coffee Average Median Household Income: {median_income_blue:.2f}")
print(f"Joe Coffee Company Average Median Household Income: {median_income_joe:.2f}")
```

```
Blue Bottle Coffee stores are located in areas with higher average household income.
Blue Bottle Coffee Average Median Household Income: 139460.43
Joe Coffee Company Average Median Household Income: 123393.55
```

```
import os

# Set the folder path where your CSV files are located
folder_path = r'C:\Users\ALIENWARE\Desktop\Urban Data Informatics\PSet 02'

# Create an empty list to store dataframes
dataframes = []

# Loop through all the files in the folder
for file in os.listdir(folder_path):
    if file.endswith('.csv'):
        file_path = os.path.join(folder_path, file)
        df = pd.read_csv(file_path)
        dataframes.append(df)

# Combine all the dataframes into one
combined_df = pd.concat(dataframes, ignore_index=True)

# Save the combined DataFrame to a new CSV
combined_df.to_csv('Coffee_Chain_Manhattan_data.csv', index=False)

print("CSV files combined successfully!")
```